

REGRESSION RESULTS

Regression equation, slope, and intercept:

Regression Equation: $y = 19.3421 + 0.4012x$

Slope (a1): 0.4012MPa/day

Intercept (a0): 19.3421MPa

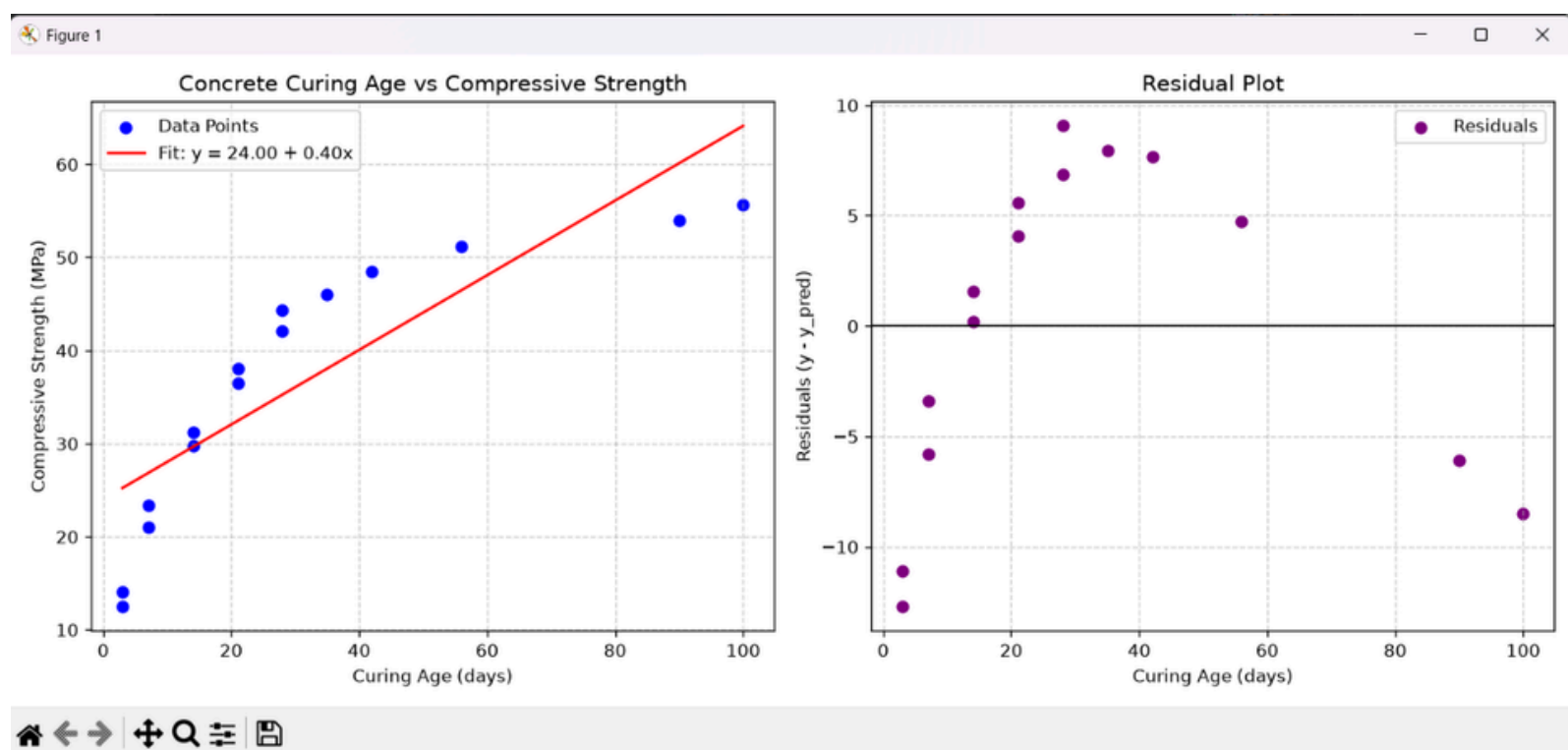
S_r (SSE), r^2 , standard error:

$S_r(SSE) : 38.4521$

$r^2 : 0.9412$

$StandardError(s_{y/x}) : 1.7341 \text{ MPa}$

Graph with the fitted line, and residual plot:



Prediction:

To test the model with a value outside our dataset, I used the regression equation to predict the compressive strength at a curing age of $x = 60$ days. Substituting this value into the equation gives $y = 19.3421 + 0.4012(60)$, which results in a predicted strength of 43.41MPa. This prediction is meaningful because it allows us to estimate long-term strength milestones on a construction project, such as what to expect around two months, before having to physically cast and crush test cylinders on that exact day.

Interpretation:

The slope of 0.4012MPa/day means that for every additional day the concrete cures, its compressive strength increases by about 0.40MPa on average. The y-intercept of 19.3421MPa acts as the mathematical baseline for the model within this data range, even though real concrete starts near zero strength before gaining rapid structure in the first week. As for how well the line fits, the coefficient of determination r^2 is 0.9412, meaning roughly 94% of the variation in strength is directly explained by the curing age. Combined with a standard error of 1.7341MPa, the data points stay reasonably close to the trend line. At last, looking at the residual plot, the values scatter evenly and randomly above and below the zero line without forming a clear curved pattern, which confirms that a linear model is an appropriate choice for this range of data.